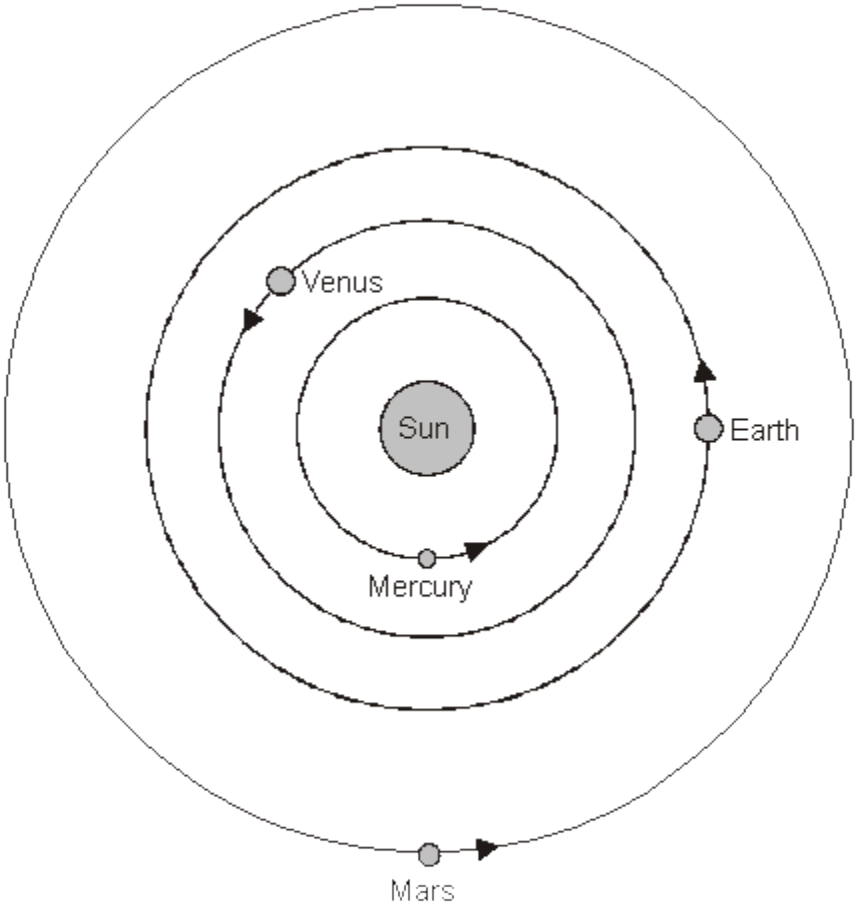
1. The table below shows information about four planets.

planet	time taken to orbit the Sun (Earth years)	distance from the Sun (million km)
Mercury	0.25	60
Venus	0.5	108
Earth	1.0	150
Mars	2.0	228

The diagram below shows the orbits of the Earth, Mercury, Venus and Mars, and their position at one particular time.
The arrows show the direction in which the planets move.



not to scale

(a) Show the position of each planet six months later by drawing a letter X on the orbit of each planet.

2 marks

(b) Use the information in the table to calculate the largest and smallest distance between the Earth and Venus.

closest ..... million km

1 mark

furthest ..... million km

1 mark

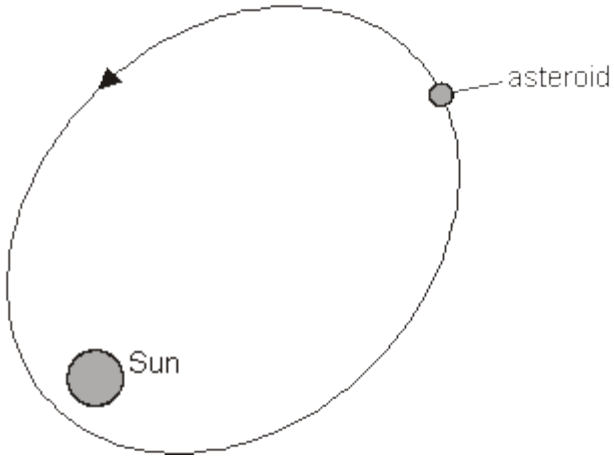
(c) The speed of light is 300 000 km/second.
Calculate how long light takes to reach the Earth from the Sun.

.....

..... s

1 mark

(d) The diagram below shows the path of an asteroid around the Sun.



not to scale

(i) **On the path of the asteroid**, draw a letter S to show the position where the asteroid is travelling the slowest.

On the path of the asteroid, draw a letter F to show the position where the asteroid is travelling the fastest.

1 mark

(ii) Explain why the speed of the asteroid changes.

.....

.....

1 mark
maximum 7 marks

2. (a) Satellites orbiting the Earth appear as points of light in the night sky. They look similar to stars. An astronomer sees a point of light in the night sky.

(i) How can she tell that the point of light is a satellite in a low orbit and not a star?

.....

.....

1 mark

(ii) She looks through a telescope and sees another point of light which is a satellite in a geostationary orbit. How can she tell that this satellite is in a geostationary orbit?

.....

.....

1 mark

(b) The Hubble Space Telescope is a telescope used to observe the planets and stars. It is in a satellite orbiting close to the Earth.

(i) Give **one** reason why a telescope in orbit gives clear pictures.

.....

.....

1 mark

(ii) Give **one** disadvantage of an astronomical telescope in orbit round the Earth.

.....

.....

1 mark

(c) In 1994, astronauts visited The Hubble Space Telescope to repair it. Astronauts have to train themselves to move objects in a controlled way in space.

Name a pair of antagonistic muscles which has to be controlled.

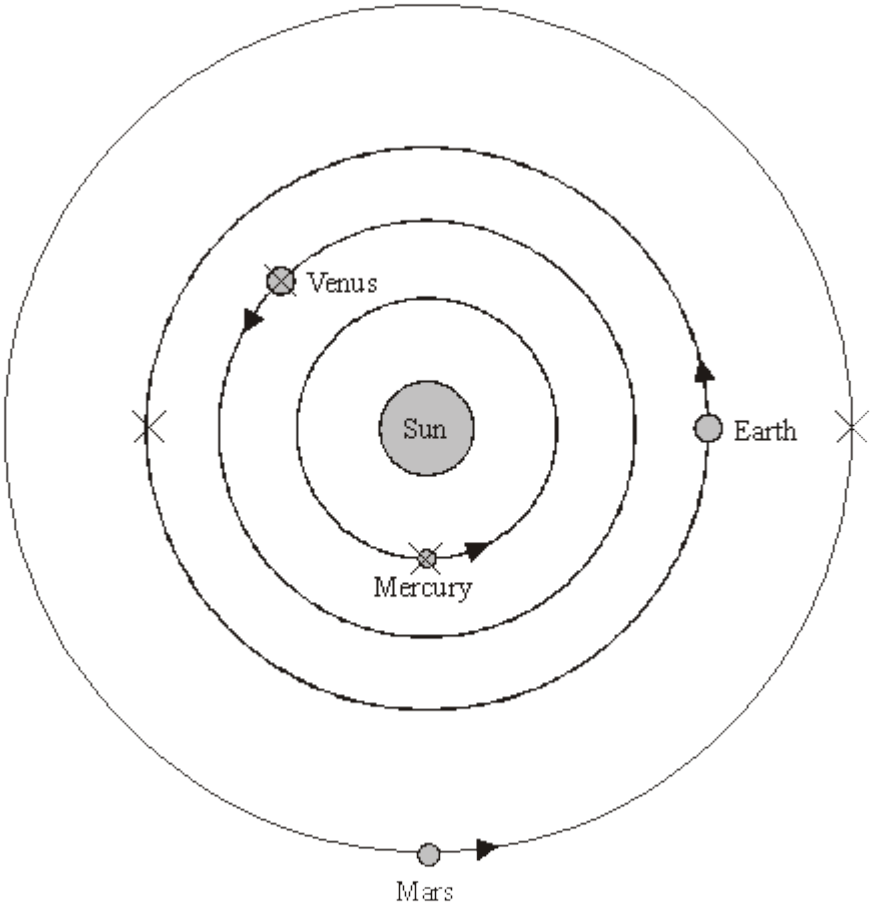
.....

1 mark

Maximum 5 marks

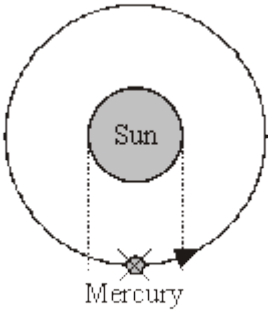
Mark schemes

1. (a)



*for all **four** correct, award two marks*
*for any **two** or **three** correct, award one mark*

accept a cross drawn that lies within the width of the Sun for each planet, e.g.



2 (L7)

(b) • 42

accept '150-108'

1 (L7)

• 258

accept '150+108'

1 (L7)

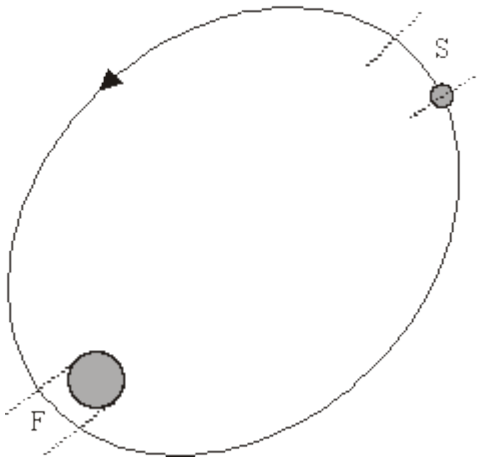
(c) 500

accept $\frac{\text{'150,000,000}}{300,000}$ ***or*** $\frac{\text{150 million'}}{300,000}$

accept '8.3 minutes'
accept 'about 8 minutes'

1 (L7)

(d) (i)



***both** answers are required for the mark*
accept an answer within the dotted lines
at either end of the ellipse

1 (L7)

(ii) attracted by the Sun's gravity

accept 'increased gravity near to Sun'

1 (L7)

[7]

2. (a) (i) it moves across the pattern of stars **or** the sky

accept 'she can see it moving' ***or*** *'it appears to move'*
or *'it moves quickly'*

*do **not** accept 'it moves'*

1

(ii) it remains above a fixed point on the Earth's surface
or it does not appear to move across the sky

accept 'it stays in the same place in the sky'
or *'it can always be seen'*

*do **not** accept 'it does not move'*

1

(b) (i) no atmosphere **or** air pollution to distort the image

*accept 'no clouds' **or** 'no light pollution'*

1

(ii) any **one** from

- difficult **or** expensive to get to for repairs **or** adjustments
- pictures have to be sent back to the Earth
- expensive to make **or** launch

1

(c) any **one** from

- biceps and triceps
- quadriceps femoris and biceps femoris

***both** muscles are required for the mark*
accept 'quads and biceps'

1

[5]